

Scaling Databases: The Shift from Vnodes to Tablets

A comparative study of database elasticity and data distribution

ScyllaDB • Vnodes and Tablets

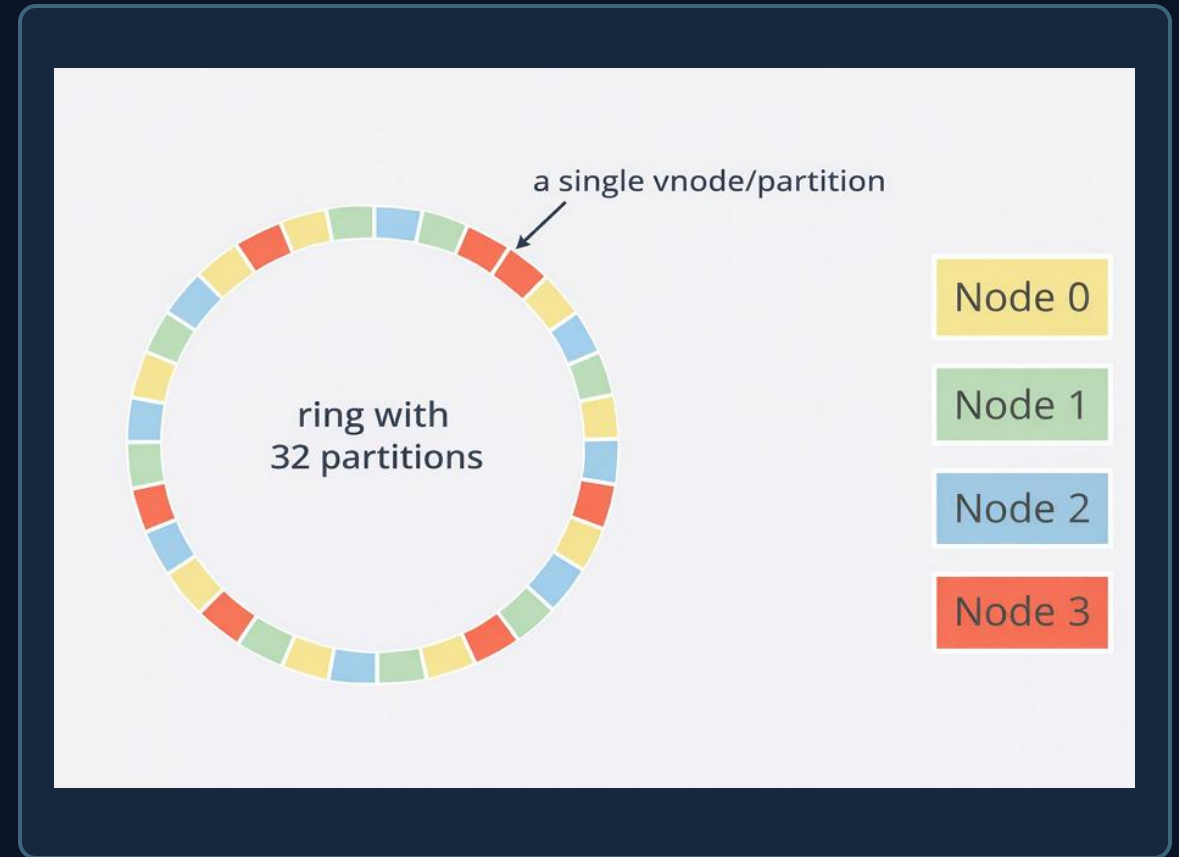
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The Core Challenge of Database Scaling

- Applications require databases to add capacity (servers) as they grow
- Adding a server means the system must move data across the network
- Clients continue to send read and write requests while data is moving
- The goal: reorganize the data without disrupting the live application

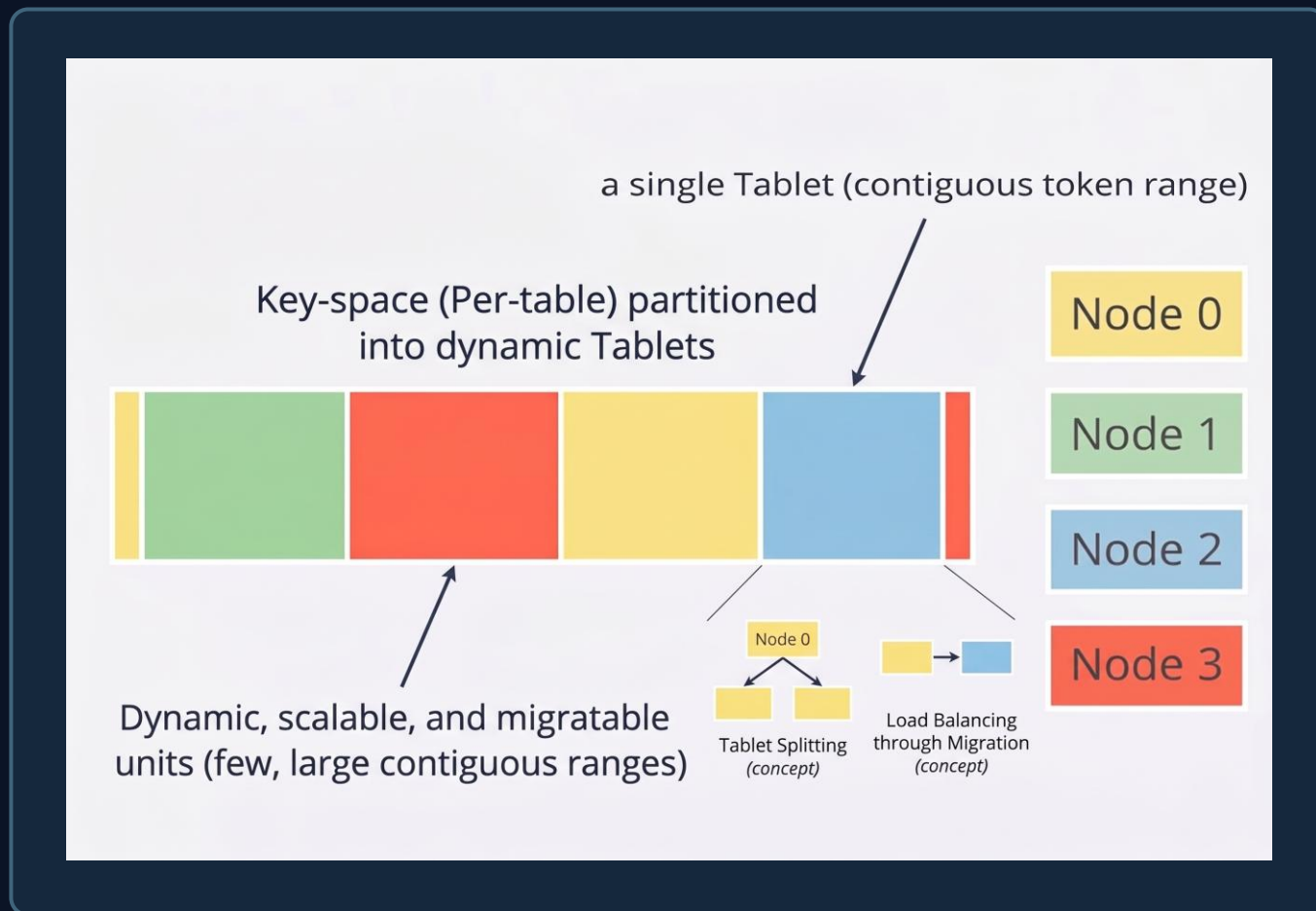
The Legacy Architecture: Vnodes

- Data is distributed using a shared, global ring of virtual nodes (vnodes)
- Files on the hard drive contain overlapping data from many different vnode ranges
- Moving data requires the system to read, filter, and rewrite these mixed files



The New Architecture: Tablets

- Each individual table gets its own dynamic, contiguous data ranges
- Physical storage files are strictly isolated to match specific tablet boundaries - you will never find data from two different tablets mixed inside the same SSTable.
- Entire data units can be cleanly migrated or deleted without filtering mixed data

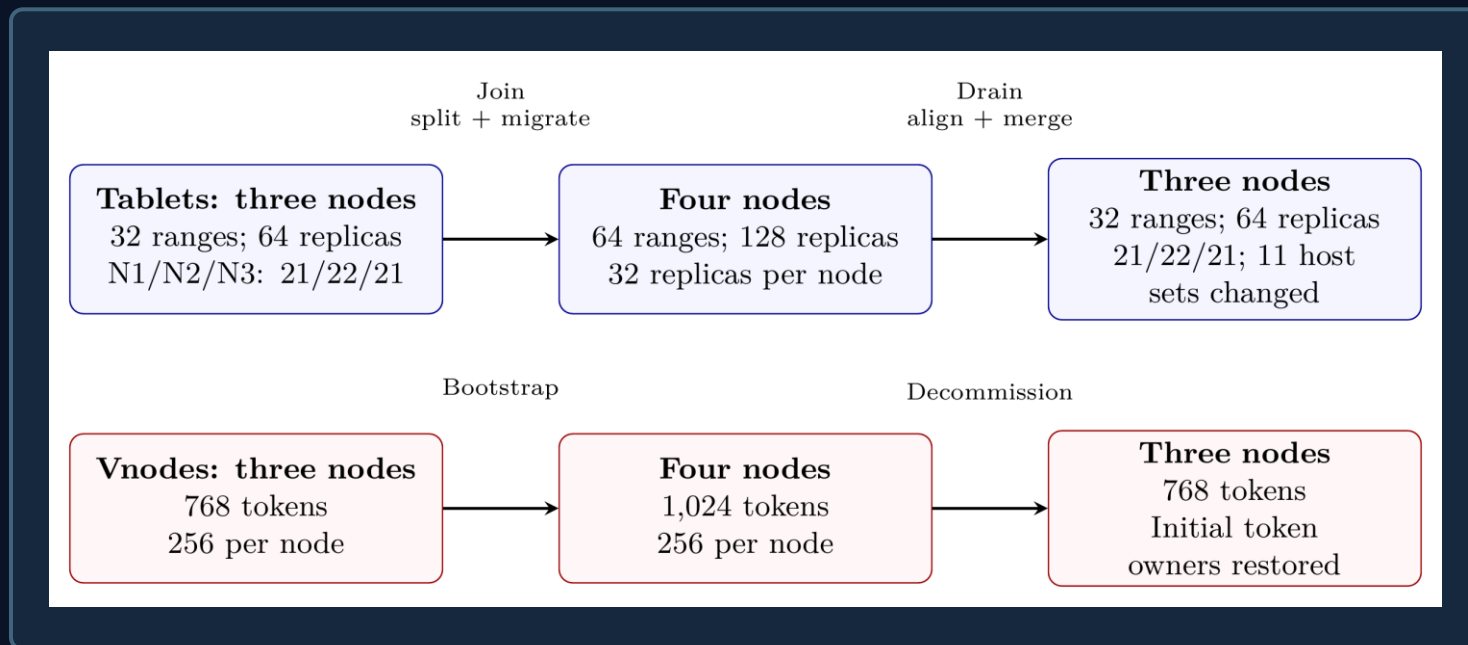


The Experiment: Lifecycle and Workload

Phase	Nodes	Duration (s)	Operations	Seed	Topology action
Baseline	3	120	60,000	20260921	None
Expansion	3 to 4	900	450,000	20260922	Add node at scheduled +120 s
Four-node	4	120	60,000	20260923	None
Contraction	4 to 3	900	450,000	20260924	Decommission at scheduled +120 s
Final three	3	120	60,000	20260925	None

- Five distinct phases: Baseline, Expansion, Four-node, Contraction, Final
- Operations during expansion and contraction ran for 15 minutes each
- Over 1 million operations were executed to ensure a rigorous stress test

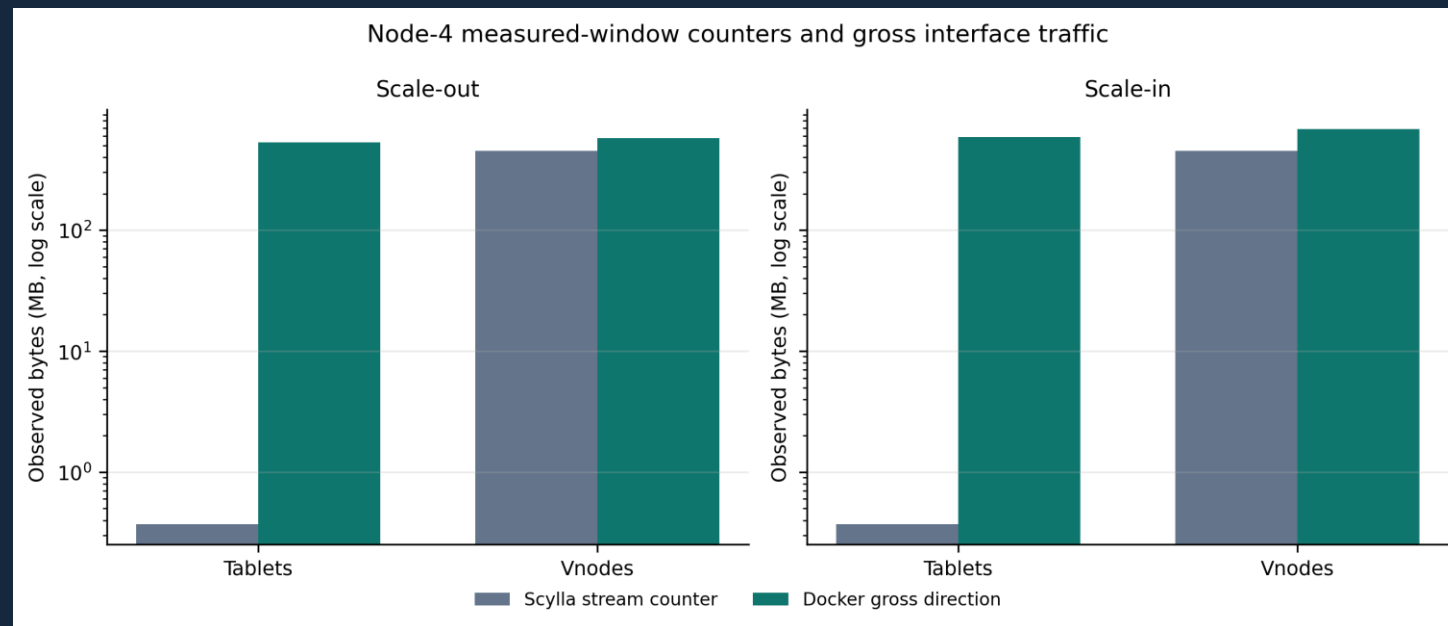
The Experiment: Expanding and Shrinking



- The test measured a full expansion (3 nodes to 4) and contraction (4 nodes to 3)
- Tablets dynamically split their 32 ranges into 64 to rebalance the cluster
- Vnodes maintained a fixed mathematical structure of 768 tokens throughout

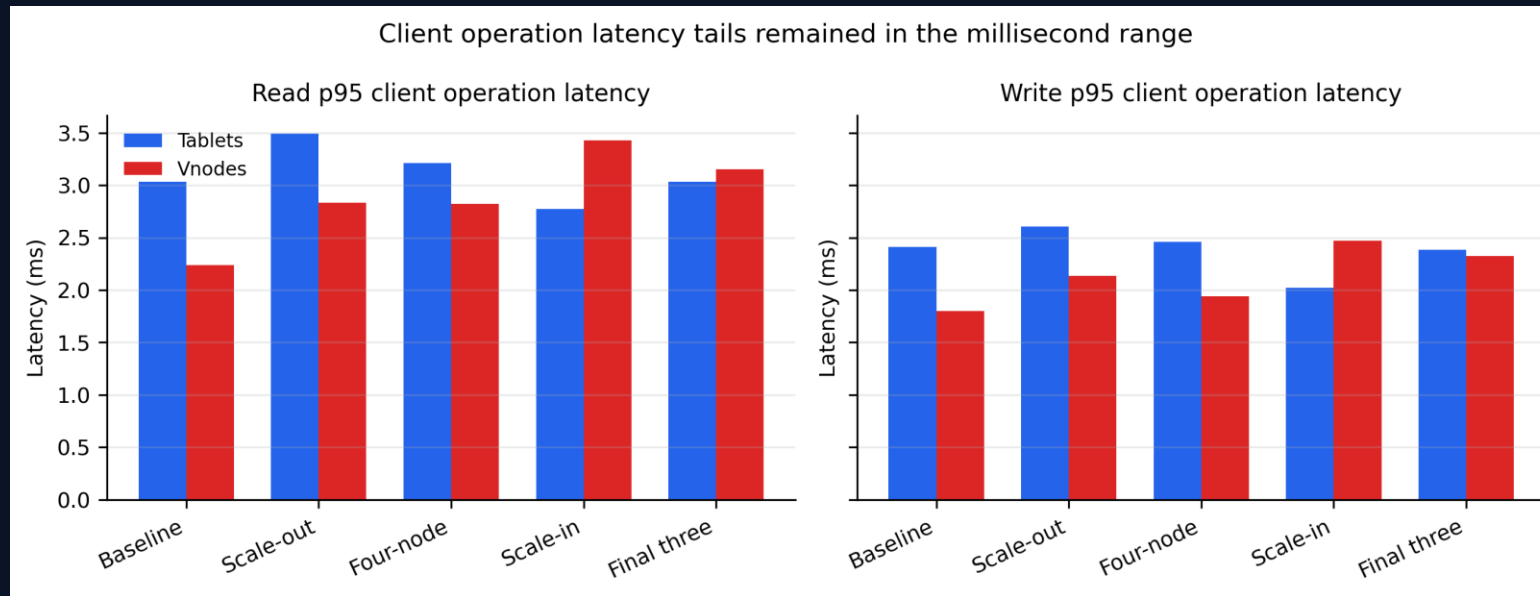
The Invisible Data Transfer

- Internal stream counters (grey) missed massive amounts of data movement for tablets
- Physical network traffic (green) revealed the true size of the data transfer
- The tablet counter is artificially low due to separate internal accounting paths



ScyllaDB built a brand new data path for moving tablets that simply wasn't plugged into the traditional stream-manager counters

The Experiment: Lifecycle and Workload



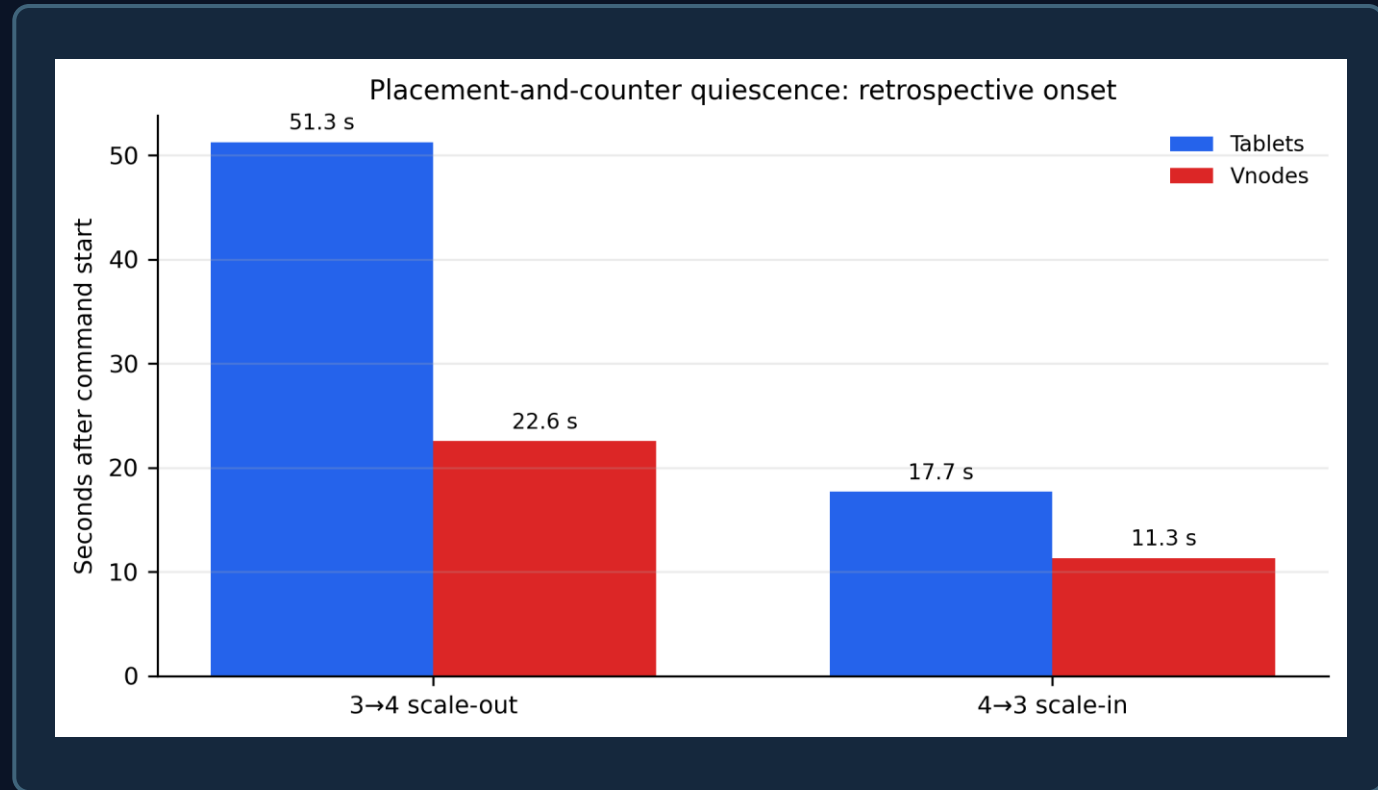
- Measures the baseline operation delay for normal user traffic
- Both systems kept standard delays successfully in the low millisecond range
- Vnodes demonstrated a faster starting baseline before scaling began



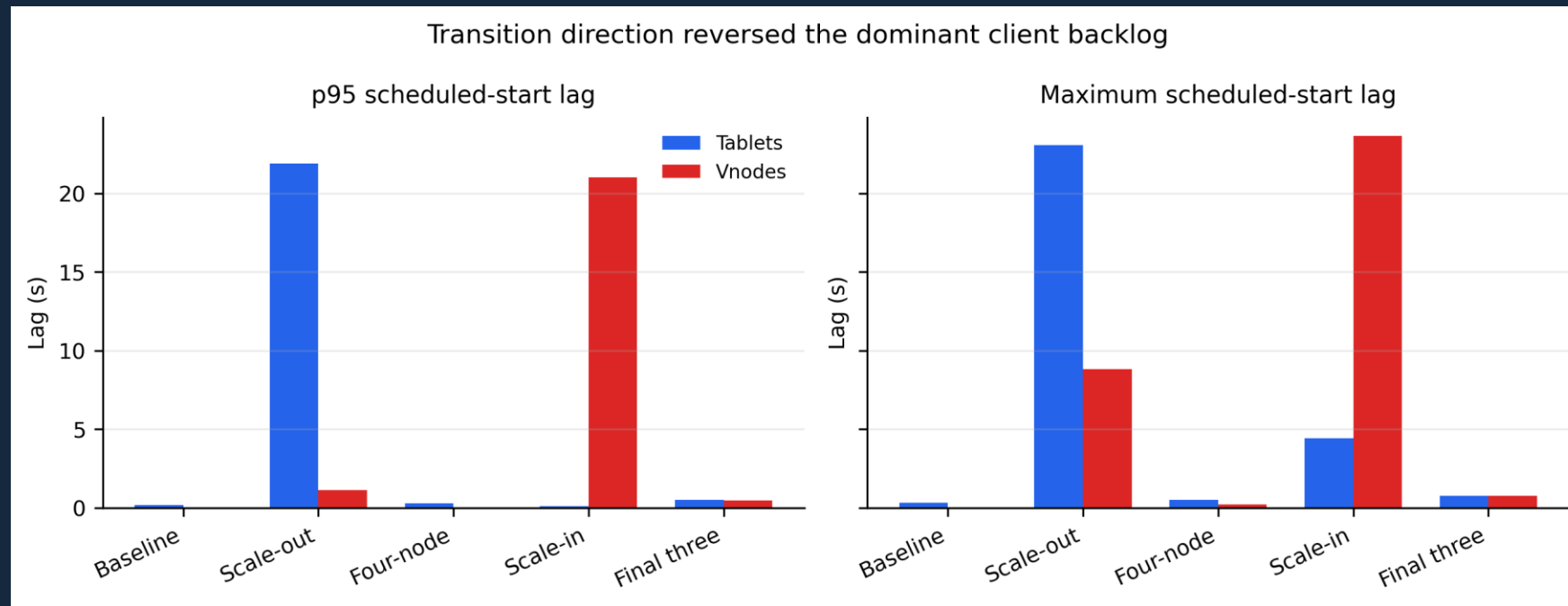
Vnodes got slightly slower every time the database finished resizing, while the tablet system restored its initial performance.

Defining "Done": Placement Quiescence

- Measures when the system's metadata officially reports that data placement is finished
- Vnodes reached this official completion milestone significantly earlier than tablets
- Is Vnode architecture really this fast during topological changes? This needs to be compared to client lag to see the full picture.



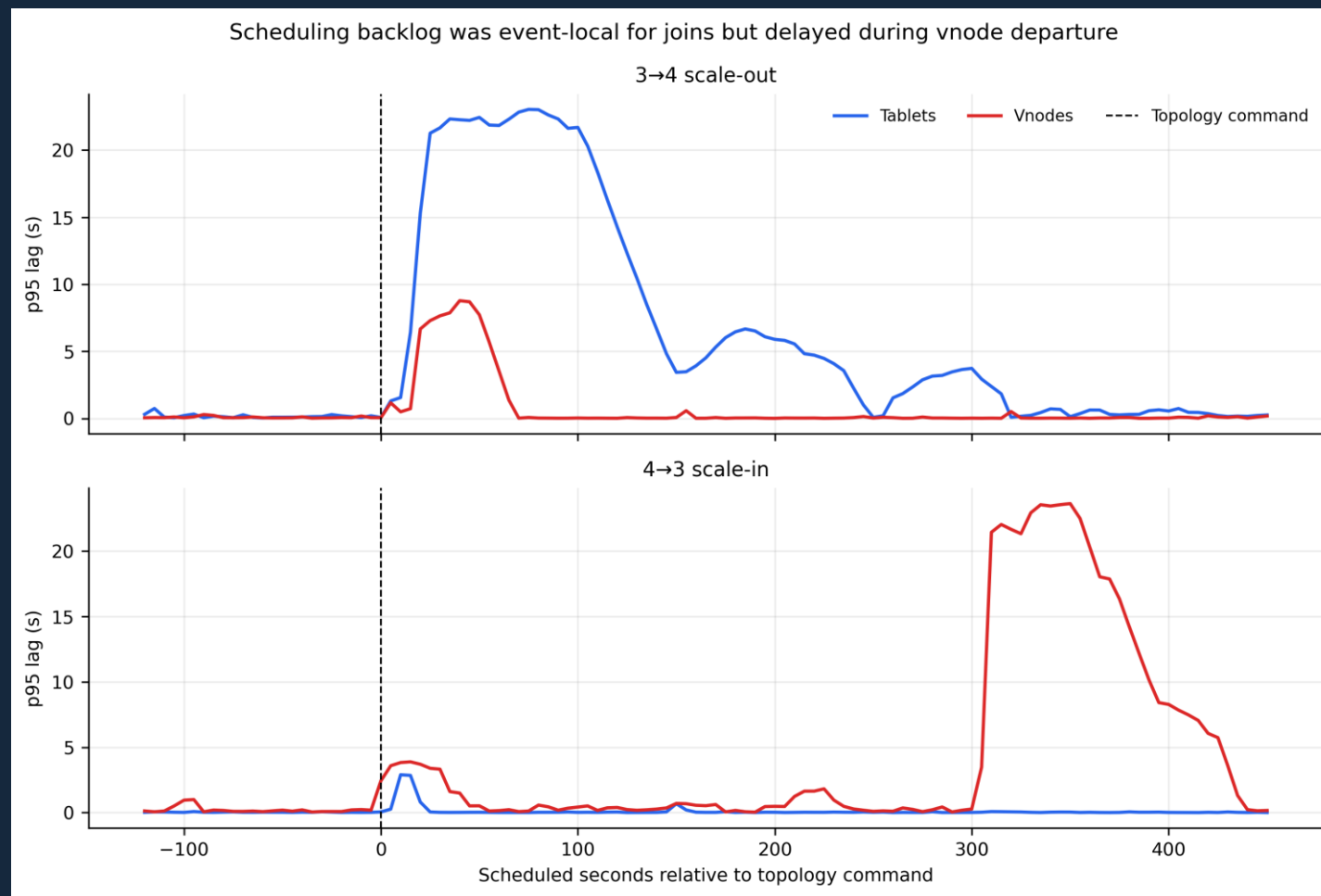
The Big Traffic Jams: Scheduling Lag



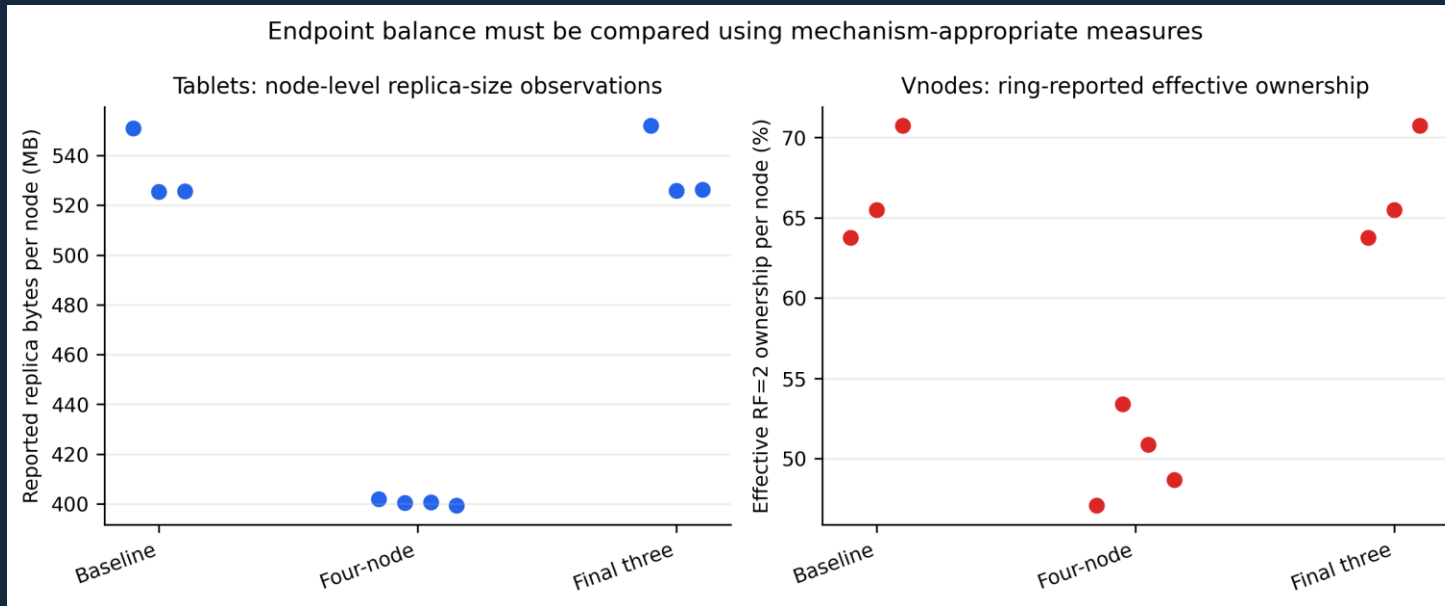
- Lag occurs when the database temporarily pauses or queues user requests
- Tablets suffered their most severe lag during cluster expansion
- Vnodes suffered their most severe lag during cluster contraction
- Both architectures experienced a service deficit exceeding 20 seconds

The 5-Minute Time Bomb

- Tablet lag (blue) occurred immediately after the topology command was given
- Vnode lag (red) spiked severely exactly five minutes after the transition started
- This five-minute delay perfectly matches an off-strategy compaction trigger in the source code



Achieving Perfect Balance



- Tablets (blue) achieved highly equal replica counts and balanced data sizes
- Vnodes (red) retained unequal effective ownership across the different nodes
- Dynamic tablet boundaries allow the system to slice and balance data perfectly

Final Takeaways

- Never trust a single metric. You have to look at the network, the hard drives, and the client delays together to understand the real story.
- Tablets had the larger client delay when adding a node, while vnodes had the larger delay when removing one.
- Vnode clients slowed down again about five minutes after node removal, when the remaining nodes began compaction.
- Tablets balanced data more evenly across the four nodes in this experiment.
- A cluster can finish changing data placement while storage work and client recovery are still underway.